

- **Rain Detection Mechanism:**
 - A rain drop sensor will detect water droplets and trigger a switching circuit that controls the plastic cover's movement.
 - A relay system will be used to actuate the cover through a gearbox DC motor.

- **Component Selection Justification:**

- **Rain Drop Sensor:** The raindrop sensors are chosen as rain detection instruments because they are easy to find and affordable.
- **Transistor (BC547):** The BC547 NPN transistor was chosen for its common usage, availability, and its ability to handle the current required by the DPDT relay (100 mA maximum current vs 10 mA needed by DPDT).
- **DPDT Relay:** DPDT relay was chosen for its availability and its ability to control the directional movement of the gearbox DC motor.
- **Gearbox DC Motor:** Gearbox DC motor provides high torque while keeping the price affordable. Its high torque enabled the plastic cover to protect the plants from rain.
- **Light Emitting Diode:** Used as an indicator to indicate state of sensor activation and idle status. LEDs were chosen because they are energy efficient and inexpensive, reducing unnecessary energy waste (only use around 10mA of current).
- **1N5822 Diode:** Used in conjunction with a limit switch to allow the motor to rotate in a limited manner and as a flywheel diode for protection. This diode was chosen due to its low price and easy to find.

- **Working Principle:**

When powered on, the red LED lights up as a power indicator. If water drops onto the rain drop sensor, the conductivity of the rain drop sensor will increase, causing the transistor to activate, triggering the relay, which in turn drives the motor to pull the plastic cover over the plants. At the same time, the green LED lights up as an indicator. The motor will continue moving the cover until it reaches the limit switch, signaling that the plants are fully covered, and the motor stops. Once the rain stops, the conductivity of the rain drop sensor will decrease, causing the transistor to deactivate, turning off the relay, causing the motor to retract the cover. The green LED turns off, indicating that the cover is retracting. The motor will keep retracting the cover until it reaches the original position and triggers the limit switch, stopping the motor.

Calculations:

Current through 500Ω resistor (to the indicator LED):

$$I = \frac{V}{R} = \frac{5}{500} = 0.01 A = 10 mA$$

3. Project Timeline:

- **Week 1-4:** Research and selection of components, schematic design, initial circuit simulation, creation of problem identification and project proposal.
- **Week 5:** Project proposal presentation and feedback from lecturer.
- **Week 5-7:** Lecturer feedback integration and project proposal refinement.
- **Week 7-8:** Circuit board construction and testing of the rain detection circuit and relay switching mechanism.
- **Week 9-11:** Integration of circuits with actuators, refinement of control logic, and testing under simulated rain conditions.
- **Week 12-13:** Final system testing, data collection on plant conditions, performance evaluation, report writing, and presentation preparation.

4. Materials/Equipment Required:

- Gearbox DC Motor
- Connecting wires
- Resistor (500Ω, 10kΩ)
- Transistor (BC547)
- Diode (1N5822)
- LED (Red, Green)
- Rain Sensor
- DPDT Relay (Double Pole Double Throw)
- Limit Switch
- Power Adapter
- Polylactic Acid (PLA) Filament
- 3D Printer
- Roll Mica Plastic
- Parafilm
- Copper/stainless steel pipe with an outer diameter of 3/8 inch

5. Potential Challenges and Mitigation Strategies:

- **Challenge:** Risk of rainwater exposure.

Mitigation: A waterproof box will be added to store the electronic components, a waterproof box will also be added to the DC motor and the gap in the box made for the cables will be sealed.

- **Challenge:** Rain sensor that stays wet even after the rain has finished.

Mitigation: The rain sensor is tilted slightly so that the sensor does not become flooded with water after the rain has finished.

- **Challenge:** Cover can be blown away by the wind.

Mitigation: designing a more sturdy cover frame and adding weights at the bottom so that the frame does not easily tip over.

6. Expected Outcomes:

- A fully functional automatic roof that moves from its base to cover the hydroponic plants when it is raining, and retracts when it stops raining.
- A functional polarity switching action using a DPDT relay, to determine which limit switch is activated for the roof.
- A comprehensive project report that includes:
 - A detailed description of the circuit design and operation.
 - A professionally drawn schematic diagram.
 - A list of all components used, with justifications for their selection.
 - A step-by-step explanation of the circuit's working principle.
 - A description of the testing procedures and results, including any observations or issues encountered.
 - An analysis of the circuit's performance and its effectiveness in protecting hydroponic plants from rain water.
 - A discussion of any challenges encountered during the project and the mitigation strategies employed.
 - Suggestions for future improvements or extensions to the project.
- An engaging project presentation that delivers the project's goals, methodology, and how the project solves the problem.